

HARSH KASANA

Full-Stack & Backend Engineer · going deep on Go & cloud infrastructure

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EDUCATION

Indian Institute of Information Technology, Nagpur

November 2022 – July 2026

B.Tech in Electronics and Communication Engineering

CGPA: 8.37

EXPERIENCE

Independent — Building in Public

June 2026 – Present

Backend & Systems Engineering

- Rebuilding core infrastructure systems from scratch in **Go** to master their internals — a Terraform-style IaC engine (**light-tf**), Git (**mygit**), and containerized HTTP services (**go-server**).
- Publishing technical breakdowns (system design, Git internals, Go vs JS architecture) as videos and threads to a growing developer audience.

Reactive Network

April 2025 – May 2026

DApp & Solidity Developer · Forward Deployed Engineer · Video Creator

- Designed cross-chain, **event-driven architecture** on Base Mainnet — autonomous contracts on a **Convex real-time backend**, monitoring positions **24/7** without bots or oracles.
- Architected and shipped **ReacDEFI**, the first production DApp on Reactive Network — **\$9,000 grant**, featured on **Chainwire**, **CryptoSlate & Blockchain Magazine**.
- As a **Forward Deployed Engineer**, shipped user integrations end-to-end and built an AI assistant (**Gemini**) that turns natural language into deployed automations.

PROJECTS

light-tf | [Code](#) | Go, IaC, gRPC, AWS SDK

2026

- A Terraform-style infrastructure-as-code engine rebuilt from scratch in **Go** — resolves resource dependency graphs with a recursive **DFS**, computing topological apply order with full **cycle detection** (visited/visiting/unvisited).
- Designed for **concurrent** resource creation via **goroutines** + **channels** and a **gRPC plugin interface** mirroring real Terraform providers; **AWS Go SDK** integration planned.

go-server | [Code](#) | Go, Docker, docker-compose, PostgreSQL

2026

- A **containerized Go HTTP API** backed by **PostgreSQL**, orchestrated with **Docker** + **docker-compose** and SQL schema initialization.
- Built from scratch to internalize production backend fundamentals — routing, database access, and container orchestration — by implementing rather than following tutorials.

mygit | [Code](#) | Go, Git internals, content-addressable storage

2026

- Reimplemented Git's core **object model** — blobs, trees, and commits in a **content-addressable store** keyed by hash.
- Built plumbing commands from first principles to internalize how Git actually stores and tracks history.

Hashtro | [Live](#) | [Code](#) | Node.js/Express, FastAPI, Next.js, Supabase

2025

- Built a **3-service microservices architecture** (Next.js, Node.js/Express API, Python/FastAPI) with Privy auth, **24h TTL caching**, rate limiting, and Postgres row-level security.
- Acquired **160 users within 48 hours** of launch behind a clean API boundary over Swiss Ephemeris computation and Google Gemini.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, C++, Rust, SQL (PostgreSQL, MySQL)

Backend & Infra: Docker, docker-compose, AWS, PostgreSQL, Redis, Node.js, Express, FastAPI, gRPC, WebSocket, Linux, Git, Turborepo

Frontend & Web3: React, Next.js, TailwindCSS, Convex, Supabase · Solidity, Reactive Smart Contracts, Foundry, wagmi/viem, cross-chain architecture

ACHIEVEMENTS

- Secured a **\$9,000 USDT grant** from Reactive Network for ReacDEFI; product launch covered by **Chainwire**, **CryptoSlate**, **BlockTelegraph**, and **Blockchain Magazine**.
- Won **1st (\$2,000)** and **2nd (\$2,000)** place at the **International Reactive Hackathon 2024**, competing against 397 hackers across 74 projects.
- Secured **two awards (\$400 + \$600)** in a US-based **Open Source Blockchain Bounty Program**; **2nd place** at the WEB3 Hackathon, IIIT SRICITY (500+ participants).